Andrew Stirling

Education

M.Sc (Thesis), Mechanical Engineering Sept. 2024 – May 2026
McGill University *Montréal, CAN*

- *GPA*: 4.0/4.0.
- *Thesis*: Ultra-wideband sensing for robotic navigation in GPS-denied environments.
- *Honors & Awards*: Ian McLachlin Prize for Entrepreneurship in Engineering, McGill Engineering Undergraduate Student Master's Award

B.Eng, Mechanical Engineering (Minor Computer Science) Sept. 2019 – May 2024
McGill University *Montréal, CAN*

- *GPA*: 3.8/4.0.
- *Honors & Awards*: Engine Design Prize, Antje Graupe Pryor Award, RISE Globalink Award, Dean's Honour List, Alexander Rutherford Scholarship.
- *Extracurriculars*: McGill Men's Volleyball Team, McGill Robotics Club, McGill Tutor, and Teaching Assistant.

Experience

M.Sc. Thesis Candidate Sept. 2024 – Present
DECAR Group, Supervised by Prof. James R. Forbes *Montréal, CAN*

- Derived a novel ML-based variational inference framework for robotic localization handling non-line-of-sight (NLOS) and multipath effects enabling improved perception and mapping performance.
- Developed a ROS package for UWB-visual-inertial SLAM with applications in GPS-denied navigation for UAVs, making use of advanced sensor fusion and machine learning techniques.

Co-Founder Sept. 2023 – Aug. 2024
OPSIS *Montréal, CAN*

- Capstone design project turned startup accepted into McGill Engine's TechAccel incubator program.
- Developed and applied a novel Kalman filtering algorithm for occlusion robust surgical tool localization in neuro-navigated surgeries.
- Assisted in the mechanical design of a multi-camera fixture for attachment to surgical lamps.

Robotics Research Intern Jun. 2023 – Aug. 2023
University of Applied Sciences & Technology *Berlin, GER*

- Conducted extensive testing of diverse quadruped control strategies utilizing MuJoCo and validated outcomes through physical testing on the Unitree Go1 Robot, utilizing a ROS2 package tailored for low-level control.
- Assisted in the development and successful implementation of an Iterative Learning Controller (ILC), capable of tracking commanded velocities within 5 % relative error.
- Co-authored paper detailing control framework accepted and presented at the European Control Conference (ECC).

Publications Under Review

[1] **Andrew Stirling**, Mykola Lukashchuk, et al. "Gaussian Variational Inference with Non-Gaussian Factors for State Estimation: A UWB Localization Case Study". In: *IEEE Robotics and Automation Letters* (Nov. 2025).

Publications

[2] Manuel Weiss, **Andrew Stirling**, et al. "Achieving Velocity Tracking Despite Model Uncertainty for a Quadruped Robot with a PD-ILC Controller". In: *2024 European Control Conference (ECC)*. June 2024, pp. 134–140.

Skills

Programming: Python, C/C^{++}, Java, MATLAB, OCAML, VBA, LaTeX, MIPS Assembly.
Libraries: NumPy, SciPy, Pandas, Matplotlib, SciKit Learn, Mosek, CVXPy.
Software: Docker, Gazebo, MuJoCo, Git, SolidWorks, ABAQUS, MasterCAM, Excel, MS PowerBI, CURA.
Operating Systems: Linux, Windows, ROS 1/2
Languages: **English** (Fluent), **French** (Advanced)